MIL-R-55154B(EL)

24 April 1972

Superseding
MIL-R-55154A(EL)
13 February 1963

MILITARY SPECIFICATION

RADIO SET CONTROL GROUP AN/GRA-39( ) (UNITS OF)

1. SCOPE

1.1 This specification covers the following units of one type of radio set remote control designated Radio Set Control Group AN/GRA-39( ):

Control, Radio Set C-2328( )/GRA-39
Control, Radio Set C-2329( )/GRA-39
Bag, Cotton Duck CW-598( )/GRA-39

Control C-2328( )/GRA-39 is used to operate certain radio sets remotely, and Control C-2329( )/GRA-39 is used to transmit and receive voice communications locally through certain radio sets. Canvas Carrying Bag CW-598( )/GRA-39 is used to store both these units and other equipment (see 6.1 and 6.8).

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bid or request for proposal form a part of the specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-S-781	Steel Strapping, Flat
PPP-B-585	Boxes, Wood, Wirebound
PPP-B-601	Boxes, Wood, Cleated-Plywood
PPP-B-621	Boxes, Wood, Nailed and Locked-Corner
PPP-B-636	Boxes, Fiberboard
PPP-F-320	Fiberboard, corrugated, and solid, sheet stock (container grade) and cut shapes
PPP-P-291	Paperboard, wrapping, cushioning
PPP-T-45	Tape, gummed, paper, reinforced and plain, for sealing and securing

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between 10 and 125 MM. Spurious emissions shall be reduced at least 60 db below with a level

FEDERAL

PPP-T-76	Tape, Pressure-Sensitive Adhesive Paper (For Carton Sealing)
PPP-T-97	Tape; Pressure-Sensitive Adhesive, Filament Reinforced

MILITARY

MIL-P-116	Preservation, Methods of
MIL-B-117	Bags, Interior packaging
MIL-I-6869	Impregnants for Aluminum and Magnesium
MIL-P-11268	Parts, Materials and Processes Used in Electronic Equipment
MIL-L-12606	Loudspeaker, Permanent Magnet
MIL-M-13231	Marking of Electronic Items
MIL-F-14072	Finishes for Ground Signal Equipment

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for shipment and storage
MIL-STD-147	Palletized unit loads C 40" x 48" 4-way partial and 4-way pallets
MIL-STD-252	Wired Equipment Classification of Visual and Mechanical Defects
MIL-STD-276	Impregnation of Porous Nonferrous Metal Castings
MIL-STD-454	Standard General Requirements for Electronic Equipment
MIL-STD-461	Electromagnetic Interference Characteristics Requirements
MIL-STD-462	Electromagnetic Interference Characteristics, Measurement of
MIL-STD-810	Environmental Test Methods

DRAWINGS

SIGNAL CORPS

SC-DL-456200	Control, Radio Set C-2329( )/GRA-39
SC-DL-456280	Control, Radio Set C-2328( )/GRA-39

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DRAWINGS (Contd)

SIGNAL CORPS

SC-DL-456290	Bag, Cotton Duck CW-598( )/GRA-39
SM-C-456331	Loudspeaker, Permanent Magnet

(Copies of specifications, standards and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the Contracting Officer. Both the title and number or symbol should be stipulated when requesting copies.)

2.2 Other publications.- The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids (or request for proposal) shall apply.

3. REQUIREMENTS

3.1 First article samples.- The contractor shall furnish five (5) pre-production samples of the units for approval, if required by the invitation for bids and contract (see 4.3 and 6.2c).

3.2 Construction.- The items covered by this specification shall be constructed in accordance with the applicable drawings listed below:

SC-DL-456280	Control, Radio Set C-2328( )/GRA-39
SC-DL-456200	Control, Radio Set C-2329( )/GRA-39
SC-DL-456290	Bag, Cotton Duck CW-598( )/GRA-39

3.3 Parts, materials and processes; general.- In addition to the requirements of this specification, the requirements of MIL-P-11268, including the selection requirements therein, shall apply, unless otherwise specified.

3.4 Finish, protective.- Units shall be finished in accordance with Specification MIL-F-14072 and the equipment drawings (see 4.4 and 6.3).

3.5 Marking.-

3.5.1 Conformance.- Marking shall conform to Specification MIL-M-13231 (see 4.4). Front panel marking shall be Group I as described in the specification.

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3.5.2 Serial numbers.- The following items shall be serial numbered:

Control, Radio Set C-2328( )/GRA-39
Control, Radio Set C-2329( )/GRA-39

3.6 Die castings (see 4.4).-

3.6.1 Quality of die castings.- Quality of die castings shall be as follows:

a. Surface conditions shall be clean, dry and free from chips.

b. Cast surface finish shall be of a painting grade which allows some streaks and chill marks which can be covered and hidden by paint.

c. Flash must be removed by grinding or filing.

d. The casting, after impregnation, shall be air tight, and no leakage shall be allowed when tested in accordance with 4.6.7 and 4.9.

e. Flatness shall be as specified in the Government-furnished drawings.

f. Dimensions shall be as specified in the Government-furnished drawings.

3.6.2 Impregnation of aluminum castings.- All exterior aluminum castings shall be impregnated in accordance with Standard MIL-STD-276, using Type II impregnating material in accordance with Specification MIL-I-6869. Castings which have been impregnated shall be marked "IMP". The impregnation shall be to correct general seepage only and shall not be used to correct poor foundry techniques or significant porosity (see 4.4).

3.7 Service conditions.- The units shall meet the following service conditions:

3.7.1 Temperature.- The units shall meet the following requirements when subjected to temperature test procedures as shown in Method 501 and 502 of MIL-STD-810.

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a. Low.- During testing as specified in 4.6.1.1, the units shall meet the following requirements:

Paragraph	Measurement	Performance
4.8.2	Interphone circuit frequency response and distortion	3.8.2
4.8.3	Power amplifier frequency response and distortion	3.8.3
4.8.4	Microphone circuit freq. response	3.8.4
4.8.5	Selectivity	3.8.5
4.8.6	Tone generator output	3.8.6

After testing, the units shall meet full specification performance for these measurements.

b. High.- During testing as specified in 4.6.1.2, the units shall meet the following requirements:

Paragraph	Measurement	Performance
4.8.2	Interphone circuit frequency response and distortion	3.8.2
4.8.3	Power amplifier frequency response and distortion	3.8.3
4.8.4	Microphone circuit frequency response	3.8.4
4.8.5	Selectivity	3.8.5
4.8.6	Tone generator output	3.8.6

After testing, the units shall meet full specification performance for these measurements.

3.7.2 Humidity (moisture-resistance).- During testing as specified in 4.6.2, the units shall meet the following requirements:

Paragraph	Measurement	Performance
4.8.2	Interphone circuit frequency response and distortion	3.8.2
4.8.3	Power amplifier frequency response and distortion	3.8.3
4.8.4	Microphone circuit freq. response	3.8.4
4.8.5	Selectivity	3.8.5
4.8.6	Tone generator output	3.8.6

After testing, the units shall meet full specification performance for these measurements.

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3.7.3 Altitude (elevation).- The units shall meet full specification performance for the following both during and after testing as specified in 4.6.3:

Paragraph	Measurement	Performance
4.8.7	Operational Inspection	3.12

3.7.4 Immersion.- The units shall show no evidence of leakage after it is tested as specified in 4.6.7.

3.7.5 Vibration, bounce and shock.-

3.7.5.1 Vibration internal (see 4.6.4).- The amplitude of vibration of any part, subassembly, or structural member of the units shall not exceed twice the amplitude of the vibration applied to the units at any frequency between 10 and 55 Hertz. After being tested in accordance with paragraph 4.6.4 there shall be no physical damage and the units shall meet the performance requirements of paragraph 3.12.

3.7.5.2 Bounce and shock.- The units shall be capable of meeting the requirements of Table I.

TABLE I - Bounce and Shock

Paragraph Heading	Paragraph	Performance after test
Bounce	4.6.5	Specified performance. No physical damage.
Shock; drop	4.6.6.a	Operable (Note A). Any physical damage shall be minor only as classified in MIL-STD-252.
Shock; bench-handling	4.6.6.b	Specified performance. No physical damage.

NOTE A: After testing, the units shall meet the requirements of paragraph 3.12.

3.7.6 Fungus.- Units and assemblies shall show no evidence of viable fungus or corrosion* when subjected to 28 days of fungus test in accordance with paragraph 4.6.8.

* Corrosion is defined as any visible degradation that can be attributed to flaky, pitted, blistered, or otherwise loosened finish or metal surface.

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3.7.7 Salt fog.- The units and assemblies shall be exposed, in their operating configuration, to 48 hours of salt fog in accordance with paragraph 4.6.9, and shall show no evidence of corrosion as evidenced by any visible degradation that can be attributed to flaky, pitted, blistered, or otherwise loosened finish or metal surface.

3.8 Electrical characteristics.-

3.8.1 Input voltage.- Nominal input voltage shall be 9.0 volts. The units shall be capable of performing the transmitter keying function with the DC input potential reduced to 6.6 volts (see 4.8.6).

3.8.2 Interphone circuit frequency response and distortion.- The overall frequency response of the interphone circuit shall not deviate more than +2.0 to -6.0 db from the reference level of 1000 Hertz. The overall distortion shall not exceed 10% over the audio range of 300 to 3000 Hertz (see 4.8.2).

3.8.3 Power amplifier frequency response and distortion.- The overall audio response of the power amplifier circuit shall be within the following limits. The overall distortion over the frequency range shall not exceed 10% (see 4.8.3).

Frequency Range in Hertz	Deviation from 1000 Hertz (in db)
300 to 2000	+1 to -3
2000 to 2500	0 to -6
2500 to 3000	0 to -10

3.8.4 Microphone circuit frequency response.- The overall audio response of the microphone circuit shall be within the following limits (see 4.8.4):

Frequency Range in Hertz	Deviation from 1000 Hertz (in db)
300 to 2000	+1 to -3
2000 to 2500	0 to -6
2500 to 3000	0 to -10

3.8.5 Selectivity (keying relay circuit).- The selectivity of the keying relay circuit shall be within the following limits (see 4.8.5):

Frequency	Sensitivity
3860 Hertz	.03 to .20 volt
3940 Hertz	.03 to .20 volt

Maximum sensitivity of the keying relay circuit shall be .03 volt or greater at the resonant frequency. The resonant frequency shall be within plus or minus 40 Hertz of 3900 Hertz (reference frequency).

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3.8.6 Tone generator output.- The resonant frequency of the tone generator shall be within plus or minus 25 Hertz of 3900 Hertz. Output level shall be 0.3 volt or greater when measured with battery voltage at 9.0 volts and 0.2 volt or greater when the battery voltage is at 6.6 volts (see 4.8.6).

3.8.7 Loudspeaker.- The loudspeaker shall meet the performance requirements of Specification MIL-L-12606, except that the loudspeaker used in Control C-2328( )/GRA-39 shall be three inches and the frequency response of the loudspeaker shall meet the requirements of USAECOM Drawing SM-C-456331.

3.9 Bag, Cotton Duck CW-598( )/GRA-39.-

3.9.1 Fabric and thread.- Fabric and thread shall be pre-shrunk or allowance shall be made for shrinkage in order to provide for satisfactory fit of finished items both before and after they are soaked in tap water at normal room temperature for 24 hours and then dried thoroughly at normal room temperature (see 4.14).

3.10 Suppression of radio interference.- The units shall meet the case and cable radiation leakage requirements specified in MIL-STD-461 for class I equipment. Tests and test methods shall be RE02 as specified in MIL-STD-462 (see 4.10).

3.11 500 hour operational requirement.- The Radio Set Control Group AN/GRA-39( ) shall operate continuously for a period of 500 hours at an ambient temperature of 70°F, with an input potential of 9 volts dc, and with no replacement of parts. At the conclusion of the 500 hours of operation, Radio Set Control Group AN/GRA-39( ) shall meet the requirements specified in 3.12 (see 4.11).

3.12 Operational inspection.- The units shall be energized and subjected to the operational test in paragraph 4.8.7 to insure qualitatively the proper functioning of the equipment, including all operating controls, to determine that the unit is operable prior to packaging.

3.13 Preconditioning (see 4.7).-

3.13.1 Bounce preconditioning.- The units shall be capable of meeting the requirements herein without subsequent processing after subjection to bounce preconditioning of 4.7.1.

3.13.2 Electrical preconditioning.- Each Radio Set Control Group AN/GRA-39 on order shall be subjected to a burn-in operation of 48 hours, at 9.0 volts dc input. The last 24 hours of the 48 hour burn-in shall be failure-free. In case of failure, the units must be repaired and run repeated. Radio Set Control Group AN/GRA-39 shall meet requirements specified herein, without subsequent processing after subjection to 4.7.2.

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3.14 Technical leterature and running spare parts.- Technical literature and running spare parts shall be furnished if specified in the contract. Running spare parts shall be identical to corresponding parts in the equipment furnished on the order (see 6.2).

3.15 Interchangeability.- Like units, assemblies, subassemblies and replaceable parts shall conform to requirement 7 of MIL-STD-454 (see 4.12).

3.16 Air-seal.- The equipment case shall provide means for enabling performance of the air-seal test. When tested as specified in 4.11, the decrease in vacuum shall not exceed 0.01 pounds per square inch, gauge, per minute (see 4.9).

3.17 Workmanship.- Radio Set Control Group AN/GRA-39( ) shall be manufactured and assembled in accordance with requirement 9 of MIL-STD-454 and the applicable portions of the following paragraphs herein:

a. In MIL-P-11268:

General requirements for plastic material and plastic parts.
Wiring and cabling, including:
- Slack.
- Protection.
- Clearance.
- Splicing.
- Connections, general.
- Grounding, general.
- Shielding on wire and cable.

Soldering and brazing.
Cleaning of units.
General requirements for securing of parts.
Process for welding.
Cleaning after welding.
Controls.
Self-tapping screws.
Cleaning prior to welding.

b. In this specification:

3.4	Finish
3.5	Marking
3.6	Casting impregnation and quality

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the government. The government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.2 Classification of inspection.- Inspection shall be classified as follows:

a. First Article inspection (does not include preparation for delivery (see 4.3)).

b. Inspection covered by subsidiary documents (see 4.4).

c. Quality conformance inspection.

(1) Quality conformance inspection of units before preparation for delivery (see 4.5).

(2) Quality conformance inspection of preparation for delivery (see 4.15).

4.3 First Article inspection.- This inspection shall consist of the inspection specified in the subsidiary documents covering the items listed in 4.4; and the inspection specified for Group A, Group B, and Group C (see Tables III, IV and V respectively). Order of environmental tests shall be as specified in Table II. Prior to inspection, the First Article Samples shall be subjected to the Bounce and Electrical Preconditioning of 4.7.

TABLE II - Order of Environmental Tests

Inspection	Number of Samples				
	Unit 1	Unit 2	Unit 3	Unit 4	Unit 5
Non-damaging					
Temperature					
Low (3.7.1) (4.6.1.1)	1				
High (3.7.1) (4.6.1.2)	2				
Altitude (3.7.3) (4.6.3)	3				
Shrinkage, Bag (3.9.1)(4.14)	4				
Suppression of Radio Interference (3.10) (4.10)	5				1
500 hour operational (3.11)(4.11)				1	
Leakage (Immersion)(3.7.4)(4.6.7)					
Potentially damaging					
Humidity (moisture-resistance) (3.7.2) (4.6.2)		1			
Fungus (3.7.6) (4.6.8)			1		
Salt Fog (Note 3)(3.7.7)(4.6.9)	6				
Vibration (3.7.5.1) (4.6.4)				2	
Bounce (3.7.5.2) (4.6.5)				3	
Shock, bench-handling (3.7.5.2) (4.6.6b)				4	
Shock, drop (3.7.5.2) (4.6.6a)				5	

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NOTE 1: Other First Article inspection may precede, follow, or be interspersed between the non-damaging tests.

NOTE 2: The order on non-damaging tests and the choice of which units are subjected to the non-damaging tests may be varied if convenient, except that the vibration tests should be performed on the same unit used for the bounce and shock tests.

NOTE 3: The units shall be thoroughly washed, cleaned, dried, and refurbished, if necessary, before proceeding with subsequent tests.

4.4 Inspection covered by subsidiary documents as part of the inspection of units before preparation for delivery:

ITEM	WHERE REQUIRED
Parts, Materials and Processes used in Electronic Equipment	3.3
Finish	3.4
Marking	3.5
Casting impregnation and quality	3.6
Loudspeaker, permanent	3.8.7
Suppression of radio interference	3.10

4.5 Quality conformance inspection of units before preparation for delivery.- Each unit produced under this specification shall be first subjected to bounce preconditioning (see 3.13.1), and then to the 24 hour electrical preconditioning burn-in (see 3.13.2) prior to selection of samples for inspection. The contractor shall perform the inspection specified in 4.4 and 4.5.1 through 4.5.3.2. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements. Failure to meet any requirement as a result of preconditioning shall be classified as a defect and recorded as such. The government will review and evaluate the contractor's inspection procedures and examine the contractor's inspection records. In addition, the government - at its discretion - may perform all or any part of the specified inspection to verify the contractor's compliance with specified requirements (see 6.9). Test equipment for government verification inspection shall be made available by the contractor.

4.5.1 Group A inspection.- Each unit of each lot of equipment shall be inspected for conformance to all the inspection and test requirements of Table III. Each lot will be subject to verification, utilizing the procedures of MIL-STD-105 using the general inspection levels and the AQL's indicated in Table III. Group A inspection shall be performed in any order which is satisfactory to the government, except that the air-seal test (4.9) shall be next to last and the operational inspection (4.8.7) shall be last (see 6.7).

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TABLE III - Group A Inspection

Inspection	Req Para	Insp Para	AQL Major	AQL Minor
Visual and Mechanical				
Control, Radio Set C2328/GRA-39( )				
Printed Circuit Boards	3.17	4.13	.65%	2.5%
SM-D-334403	3.17	4.13	.65%	2.5%
SM-D-456286	3.17	4.13	.65%	2.5%
Chassis Assembly	3.17	4.13	.65%	2.5%
Front Panel Assembly	3.17	4.13	.65%	2.5%
Front Panel and Chassis Assy	3.17	4.13	1.0%	4.0%
Complete Control, Radio Set C-2328/GRA-39( )				
Control, Radio Set C2329/GRA-39( )				
Printed Circuit Boards	3.17	4.13	.65%	2.5%
SM-D-334399	3.17	4.13	.65%	2.5%
SM-D-456271	3.17	4.13	.65%	2.5%
Chassis Assembly	3.17	4.13	.65%	2.5%
Front Panel Assembly	3.17	4.13	.65%	2.5%
Front Panel and Chassis Assy	3.17	4.13	1.0%	4.0%
Complete Control, Radio Set C-2329/GRA-39( )				
Case and Cover	3.17	4.13	1.0%	4.0%
Bag, Cotton Duck CW-598( )/GRA-39( )	3.17	4.13	1.0%	4.0%
Battery box assembly	3.17	4.13	1.0%	4.0%
Speaker	3.17	4.13	1.0%	4.0%
Radio Set Control Group AN/GRA-39( )	3.17	4.13	1.0%	4.0%
Handset H-189( )/U	3.17	4.13	1.0%	4.0%
Sling, Carrying	3.17	4.13	1.0%	4.0%
Electrical				
Control, Radio Set C-2328/GRA-39( )				
Input voltage	3.8.1	4.8.6	1% for the group	N/A
Interphone circuit frequency				
Frequency response	3.8.2	4.8.2		
Distortion	3.8.2	4.8.2		

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TABLE III - Group A Inspection (Cont'd)

Inspection	Req Para	Insp Para	AQL Major	AQL Minor
Electrical (Cont'd)				
Power Amplifier Frequency				
Frequency response	3.8.3	4.8.3		
Distortion	3.8.3	4.8.3		
Microphone circuit frequency				
Frequency response	3.8.4	4.8.4		
Keying Relay Circuit				
Selectivity	3.8.5	4.8.5		
Tone Generator				
Output	3.8.6	4.8.6		
Control, Radio Set C-2329/GRA-39()				
Input voltage	3.8.1	4.8.6		
Interphone circuit frequency			1% for	
Frequency response	3.8.2	4.8.2	the	
Distortion	3.8.2	4.8.2	group	N/A
Power Amplifier Frequency				
Frequency response	3.8.3	4.8.3		
Distortion	3.8.3	4.8.3		
Microphone Circuit Frequency				
Frequency	3.8.4	4.8.4		
Keying Relay Circuit				
Selectivity	3.8.5	4.8.5		
Tone Generator				
Output	3.8.6	4.8.6		
Air-Seal				
Control, Radio Set C-2328/GRA-39()	3.16	4.9	1%	N/A
Control, Radio Set C-2329/GRA-39()	3.16	4.9	1%	N/A
Operational				
Control, Radio Set C-2328/GRA-39()	3.12	4.8.7	1%	N/A
Control, Radio Set C-2329/GRA-39()	3.12	4.8.7	1%	N/A

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4.5.2 Group B inspection.- This inspection (including sampling) shall conform to Table IV and to the procedures for small-sample inspection of Standard MIL-STD-105, using the special inspection levels. Group B inspection shall normally be performed on inspection lots that have passed Group A inspection and on samples selected from units that have been subjected to and met the Group A inspection.

TABLE IV - Group B Inspection

Inspection	Req Para	Test Para	AQL
Immersion			
Control, Radio Set C-2328/GRA-39( )	3.7.4	4.6.7	10%
Control, Radio Set C-2329/GRA-39( )	3.7.4	4.6.7	10%
Interchangeability			
Control, Radio Set C-2328/GRA-39( )	3.15	4.12	6.5%
Control, Radio Set C-2329/GRA-39( )	3.15	4.12	6.5%

4.5.2.1 Group B sampling plans.- The Group B sampling plans, for the acceptable quality levels (AQL) listed in Table IV, shall be as follows:

AQL	Inspection Level
10%	S-4
6.5%	S-4

4.5.2.2 Order of inspection within Group B.- Group B inspection shall be performed in any order which is satisfactory to the government.

4.5.3 Group C inspection.- This inspection shall be as listed in Table V, and shall normally be performed on sample units that have been subjected to and met Group A and Group B inspection. Order of test shall conform to Table II.

4.5.3.1 Sampling for inspection of equipment.- Units selected for each Group C inspection shall be selected without regard to their quality in accordance with the following:

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TABLE V - Group C Inspection

Inspection	Req Para	Insp Para
Sub-Group I		
500 hour life test	3.11	4.11
Temperature	3.7.1	4.6.1
Altitude	3.7.3	4.6.3
Vibration	3.7.5.1	4.6.4
Bounce	3.7.5.2	4.6.5
Salt Fog	3.7.7	4.6.9
Shrinkage	3.9.1	4.14
Sub-Group II		
Humidity	3.7.2	4.6.2
Shock, bench-handling	3.7.5.2	4.6.6a
Shock, drop	3.7.5.2	4.6.6b
Sub-Group III		
Fungus	3.7.6	4.6.8
Suppression of Radio Interference	3.10	4.10

4.5.3.1.1 Sub-Group I.- For this subgroup, three units from the first 50 production units shall be selected for the first group C inspection. For subsequent group C inspections, three units per month shall be selected.

4.5.3.1.2 Sub-Group II.- For this subgroup. two units from the first 50 production units shall be selected for the first group C inspection. For subsequent group C inspections, two units from each successive 1000, or two units every 4 months if less than 1000 are produced during the 4 months period, shall be selected.

4.5.3.1.3 Sub-Group III.- For this subgroup, two units from the first production lot only will be selected for the first group C inspection (not excluding First Article).

4.5.3.2 Noncompliance.- The contractor shall immediately report in writing each group C failure occurrence, including all details of the failure and characteristics affected. The contractor shall immediately investigate the cause of failure and further report the results of investigation and details of the proposed corrective action on (i) the process and materials, as applicable, and (ii) all units of product which were manufactured under the same conditions and which the government considers subject to the same failure. Reports shall be forwarded to the responsible technical activity designated in the contract through the

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Quality Assurance Representative. After corrective action has been taken, additional sample units shall be subjected to group C inspection (all inspections, or the inspections which the sample failed) at the option of the government, and group A and B inspection may be reinstituted; however, final acceptance and shipment will be withheld until the group C reinspection results have shown that the corrective action was effective.

4.5.3.3 Reinspection and disposition of conforming Group B and Group C sample units.- Unless otherwise specified, sample units which have been subjected to, and passed both Group B and Group C inspection may be accepted on the contract or order provided all damage is repaired and the sample units are resubjected to, and pass, Group A inspection.

4.6 Service conditions tests (see 3.7).-

4.6.1 Temperature test.-

4.6.1.1 Low temperature.- The units shall be tested in accordance with Method 502, Procedure 1, of MIL-STD-810. The storage temperature shall be -80°F, and the operating temperature shall be -40°F. Prior to testing, the units shall be tested and shall meet full specification performance for the measurements specified in 3.7.1a.

4.6.1.2 High temperature.- The units shall be tested in accordance with Method 501, Procedure II, of MIL-STD-810. The operating temperature shall be +150°F. Prior to testing, the units shall be tested and shall meet full specification performance for the measurements specified in 3.7.1b.

4.6.2 Humidity (moisture-resistance) test.- The units shall be tested in accordance with Method 507, Procedure II, of MIL-STD-810. Prior to testing, the units shall meet full specification performance for the measurements specified in 3.7.2. Measurements shall be made during the periods shown on the figure for the cycle. Do not move units from the humidity chamber for measurements, and do not leave power on after measurements have been completed. If any unit fails to meet the specified performance during cycling, it shall be realigned or readjusted once and the test continued. No repair or replacement of parts shall be permitted. If the unit still fails to meet the specified performance during cycling, or fails subsequently, it does not pass the tests. In addition, if the unit fails to meet specified requirements after final conditioning and adjustment, it does not pass the test. The test shall be performed with the seal broken or with the chassis removed from its enclosure.

4.6.3 Altitude (elevation) test.- The units shall be tested in accordance with Method 500, Procedure I, of MIL-STD-810. Prior to testing, the units shall meet full specification performance for the measurements specified in 3.7.3.

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4.6.4 Vibration.- The units shall be tested in accordance with Method 514, Procedure IX, Part 1, of MIL-STD-810. After testing, the units must comply with 3.7.5.1.

4.6.5 Bounce test.- The units shall be tested in accordance with Method 514, Procedure XI, Part 2, of MIL-STD-810. The units shall be energized near the end of each 1/2 hour period, and shall be tested for compliance with 4.8.7 (see 3.7.5.2).

4.6.6 Shock test (see 3.7.5.2).-

a. Drop.- The units shall be tested in accordance with Method 516, Procedure II, of MIL-STD-810.

b. Bench-handling.- The units shall be tested in accordance with Method 516, Procedure V, of MIL-STD-810.

4.6.7 Immersion test (see 3.7.4).- The units, as prepared for field transportation, shall be tested in accordance with Method 512, Procedure I, of MIL-STD-810.

4.6.8 Fungus.- The units and assemblies shall be subjected to 28 days of the fungus test specified in Method 508, Procedure I, of MIL-STD-810. For gasket or other non-hermetically sealed equipments, the parts and materials of the chassis shall be sprayed with the spore suspension. The interior of the case shall also be sprayed. The units shall be subjected to the test in this open condition (see 3.7.6).

4.6.9 Salt fog.- The units and assemblies shall be exposed in their operating configuration to salt fog in accordance with Method 509, Procedure I, of MIL-STD-810 for 48 hours (see 3.7.7).

4.7 Preconditioning (see 3.13).-

4.7.1 Bounce preconditioning.- The 2 units shall be placed in their normal operation position on the table of the Package Tester, Type 1000-SC, as made by the L.A.B. Corporation, Skaneateles, New York, or equal. The package tester, shafts in phase, shall have a speed such that it is just possible to insert a 1/32 inch thick strip of material under one corner or edge of the units to a distance of 3 inches as the units bounce. The units shall be subjected to this preconditioning for 1 minute. After bounce preconditioning, the units shall not be repaired, aligned, cleaned, or otherwise changed prior to subjection to inspection (see 3.13.1).

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4.7.2 Electrical preconditioning.- Radio Set Control Group AN/GRA-39 shall be continuously operated for a period of 48 hours at a temperature of 40 ± 5°C (95°F to 113°F) and with an input voltage of 9.0 volts dc. The timer circuit shall be set for a duty cycle of one (1) minute "Transmit", two (2) minutes "Receive", and seven (7) minutes standby. While the unit is in this operating mode, vibration shall be performed as follows: 2.2G ± 10% at any non-resonant frequency between 20 to 60 cycles per second measured at the mounting points of the unit. This vibration shall be applied for 10 minutes during each hour of unit operating time. After preconditioning, there shall be no repair, cleaning, realignment, or other changes prior to acceptance inspection (see 3.13.2).

4.8 Electrical tests (see 3.8).-

4.8.1 Standard test conditions.- All tests shall be conducted under the following conditions, except as specified for the individual tests:

a. Ambient temperature 77°F ± 5°F.

b. Ambient humidity - Normal factory ambient.

c. Ambient atmospheric pressure - Normal factory ambient.

d. Battery voltage 9.0 (nominal).

e. Telephone line impedance 600 ohms (nominal).

f. Handset receiver impedance 1000 ohms (nominal).

g. Microphone impedance 150 ohms (nominal).

h. Speaker impedance 8 ohms (nominal).

i. Radio Set output impedance 600 ohms (nominal).

j. Microphone output voltage 2.5 mv (nominal).

k. Audio response measurements shall be made at the following frequencies: 300, 1000 and 3000 Hertz.

l. A 600 ohm resistor shall be connected in series with one of the telephone leads between the local and remote control units.

4.8.2 Interphone frequency response and distortion.- Standard test conditions shall prevail. The local and remote control units shall be interconnected by means of telephone lines and the function switch on both units shall be secured in telephone position. An audio oscillator

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with an output impedance adjusted to simulate the impedance of the microphone shall be applied to pins "A" and "D" of the front panel audio connector on the local control unit. The audio output at the remote control shall be monitored with an output meter and distortion analyzer connected through a proper matching network to pins "A" and "B" of the front panel audio connector. The audio oscillator shall be adjusted to deliver a constant level of 2.5 millivolts over the audio range to the microphone input. As the frequency is varied from 300 to 3000 Hertz, the audio output level and distortion shall be measured and noted. This test shall be repeated with the audio oscillator coupled to the remote unit and the audio response measured at the local control unit. The audio output level referenced to 1000 Hertz and distortion measurements shall comply with the requirements of 3.8.2.

4.8.3 Power amplifier frequency response and distortion.- The local and remote control units shall be interconnected as in the previous test. With standard test conditions prevailing, an audio signal generator with a proper matching network shall be connected to pins "A" and "B" of the radio set coupling cable on the local control unit. The speaker on the remote unit shall be replaced with an audio output meter adjusted to the proper impedance. A distortion meter shall be connected in parallel with the output meter. The signal source simulating the radio set shall be adjusted for one milliwatt output at 1000 Hertz and the volume control on the remote control unit shall be adjusted for 300 milliwatts speaker output. With the frequency of the audio oscillator varied from 300 through 3000 Hertz, the audio output and distortion shall be measured and recorded. The audio response referenced to 1000 Hertz and distortion measurements shall meet the requirements of 3.8.3.

4.8.4 Microphone circuit frequency response (from remote unit to radio set modulator circuit).- Standard test conditions shall prevail. The local and remote control units shall be interconnected by means of telephone lines. An audio oscillator shall be coupled through a proper matching network to pins "A" and "D" (microphone input) of the front panel connector on the remote control unit. An audio frequency volt meter shall be connected to pins "A" and "D" of the radio set coupling cable on the local control unit. The audio output level shall be recorded as the frequency of the oscillator is varied from 300 through 3000 Hertz while maintaining a constant input level of 2.5 millivolts. The audio response referenced to 1000 Hertz shall comply with requirements of 3.8.4.

4.8.5 Selectivity (keying relay circuit).- An audio signal generator terminated with 600 ohms shall be coupled to the line terminals of the local control unit. A test circuit consisting of a lamp bulb of proper voltage and wattage rating in series shall be connected to pins "A" and "C" (transmitter keying circuit) of the radio set coupling cable and used to indicate closure of the keying relay. The frequency of the

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signal generator shall be set to 3860 Hertz with the aid of a frequency counter. The output of the signal generator shall be increased until an indication of relay closure is obtained. The above procedure shall be repeated with the signal generator frequency set at 3940 Hertz. The output of the signal generator shall be set at 0.03 volt and the frequency varied slowly from 3860 to 3940 Hertz. The output voltage readings shall comply with the sensitivity requirements of 3.8.5.

4.8.6 Tone generator output.- Standard test conditions shall prevail. A frequency counter and audio frequency volt meter shall be connected across the telephone lines between the local and remote control units. The volume control at the remote control unit shall be set to the minimum position. As the handset push-to-talk button at the remote control unit is depressed, the tone level and frequency shall be measured and recorded. This test shall be repeated with the DC input voltage reduced to 6.6 volts and shall comply with the requirements of 3.8.6.

4.8.7 Operational test.- Prior to packaging, completely assembled units shall be set up for remote control operation. A 600 ohm resistor shall be connected in series with one of the telephone lines and the radio set coupling cable shall be connected to a test jig. The test jig shall consist of the following circuits terminated with a five (5) pin panel type audio connector. The units under test shall be sufficiently distanced from each other so as to prevent interference and/or misleading results (3.12).

a. A lamp bulb (of proper voltage and wattage rating) in series with a battery shall be connected to pins "A" and "C". (Relay Closure Test).

b. An audio signal generator with a proper matching network and set to 1000 Hertz at one (1) milliwatt shall be connected to pins "A" and "B". (Simulated Received Audio Test).

c. A Ballentine Vacuum Tube Voltmeter shall be connected to pins "A" and "D". (Transmitter Audio Modulation Input Test).

With the equipment set up above, the following circuitry shall be checked for satisfactory operation:

(1) Duplex telephone communication between the local and remote control units.

(2) Two way (aural and visual) interphone signalling. Procedure for checking is as follows: Press and release the ringer button several times on the remote control unit. The operator at the local control unit shall aurally and visually obtain a signal. Repeat the procedure using the ringer button on the local control unit. The operator at the remote control unit shall visually and aurally obtain a signal.

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(3) From remote control unit test "Relay Closure", "Simulated Received Audio", and "Transmitter Audio Modulator Input" (minimum 2.5 millivolts with normal speaking level).

4.9 Air-seal test (see 3.16).- Remove POWER indicator light cover to break seal and to make vacuum connection. Immediately thereafter, the units, as field transported, shall be subjected to a vacuum of 1 pound per square inch, gage (1 pound per square inch less than the atmospheric pressure surrounding the equipment), applied to the interior of the case. The vacuum then shall be valved-off and the interior pressure measured during the ensuing period of 1 minute. During this 1-minute period, the decrease in vacuum shall not exceed 0.01 pound. The gauge used for measurement of the vacuum shall be of such accuracy that a difference of 0.01 pound can be determined readily.

4.10 Suppression of radio interference (see 3.10).- The units shall be subjected to the tests specified in MIL-STD-461, Table A-1-MIL-E-55301. Radio Set Control Group AN/GRA-39 shall only be tested in accordance with Test Method RE02 (Case and Cable Radiation Test) and shall comply with interference limits in Figure A-1b of MIL-STD-461. The units shall be tested with a simulated two (2) miles transmission line. The simulation can be accomplished by inserting a three hundred (300) ohm resistor in each side of the transmission line.

4.11 500 hour life test.- The local and remote control units shall be interconnected by means of telephone lines. An audio Signal Generator set at 1000 Hertz shall be connected to the local control unit through pins "A" and "B" of the radio set coupling cable connector. During "receive" condition, the audio signal shall be fed from a local control unit to the audio amplifier in the remote unit. An audio output meter shall be connected to the speaker output terminals. The signal generator output at the local control unit shall be adjusted to 300 milliwatts speaker audio output. A light indicator consisting of a lamp bulb of proper voltage and wattage rating and a battery in series connected between pins "A" and "C" on the radio set coupling cable shall be used to indicate closure of the keying relay during "transmit" condition. The timer circuit shall be set for a duty cycle of one (1) minute "transmit", two (2) minutes "receive" and seven (7) minutes standby. The units shall be operated continuously for twenty three (23) hours "on" and one (1) hour "off" (3.11).

4.12 Inspection for interchangeability.- The dimensions listed below shall be gaged or measured to determine conformance to the physical interchangeability requirement of 3.15. When a listed dimension is not within specified or design limits, it shall be considered a major defect.

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Mechanical (dimensions).- The gage points for Controls C-2328( )/GRA-39 and C2329( )/GRA-39 are listed below:

a. CASE

(1) Check overall inside dimensions, width, length and depth.

(2) Check location of locking clamps on case.

(3) Check overall outside dimensions on top and bottom lips (to insure scaling in groove).

b. CHASSIS

(1) Check overall outside dimensions of chassis width, length and height.

(2) Check location of locking clamp clips on front panel.

(3) Check width and location of gasket groove in front panel.

c. BOTTOM COVER

(1) Check width and location of gasket in cover.

(2) Check location of locking clamps clips.

4.13 Visual and mechanical inspection.- The units shall be examined for the applicable defects listed in MIL-STD-252 and Table VI. The bag shall be examined for workmanship, mechanical fit, application of specified treatment, and miscellaneous defects. Fastening devices shall be inspected for mechanical operation.

TABLE VI
Classification Visual and Mechanical Defects for Bag

Classification	Defect
Major	End clips missing or loose Broken stitches Strap missing or loose Fastening devices missing, inoperable or incorrectly attached Loose stitching Holes in material
Minor	Color not as specified Marking not as specified Stains on material

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4.14 Shrinkage test.- The carrying bag shall be inspected for conformance to 3.9.1.

4.15 Acceptance inspection of preparation for delivery.- Preparation for delivery shall be inspected in accordance with Specification MIL-P-116 to determine conformance to the requirements of section 5.

4.16 Rough handling test (preparation for delivery).- When rough handling test in accordance with Specification MIL-P-116 is required by the contract (see 6.2i), the following functional test shall be conducted to determine freedom from operational malfunction caused by the rough handling:

Paragraph	Measurement
3.12	Operational Inspection

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.- Preservation and packaging shall be level A or C as specified (see 6.2h).

5.1.1 Level A.-

5.1.1.1 Cleaning.- Radio Set Control Group AN/GRA-39( ) shall be cleaned in accordance with process C-1 of MIL-P-116.

5.1.1.2 Drying.- Radio Set Control Group AN/GRA-39( ) shall be dried in accordance with the applicable procedure of MIL-P-116.

5.1.1.3 Preservation application.- None required.

5.1.1.4 Unit packaging.- Each Radio Set Control Group AN/GRA-39( ) shall be packaged in accordance with items 1 through 4 of figure 1 of Table VII.

5.1.2 Level C.- Radio Set Control Group AN/GRA-39( ) shall be preserved and packaged in a manner that will afford adequate protection against physical and environmental damage during shipment, handling and limited intransit storage.

5.2 Packing.- Packing shall be level A, B or C as specified. Shipping containers for all levels shall be capable of stacking and supporting superimposed loads during shipment and storage without damaging the container or its contents (see 6.2h).

5.2.1 Level A.-

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5.2.1.1 Consolidation.- A quantity of Radio Set Control Group AN/GRA-39( ), packaged as specified in 5.1, shall be packed within a close-fitting fiberboard box conforming to PPP-B-636, type CF, class weather-resistant. Box closure shall be as specified in the appendix of the box specification. To facilitate palletization, fiberboard boxes shall be uniform in size and contain equal quantities of the packaged items to the greatest extent praticable.

5.2.1.2 Palletized load.- A quantity of containers, packed as specified in 5.2.1.1, shall be placed on a pallet, load type 1, conforming to MIL-STD-147. A fiberboard cap shall be employed over the load having two sides extending down the stacked load at least 12 inches to accommodate marking requirements. The cap shall be fabricated of fiberboard conforming to PPP-F-320, class weather-resistant, W5s or V3c. The load shall be "bonded" to the pallet by strapping.

5.2.1.3 Less than palletized load.- When quantities per destination are less than a pallet load, the containers packed as specified in 5.2.1.1 shall be waterproofed, with tape conforming to PPP-T-76, in accordance with the taping requirements of the appendix of the box specification. A quantity of the waterproofed containers shall be placed within a close-fitting box conforming to PPP-B-601, overseas type; PPP-B-621, style 4, class 2; or PPP-B-585, style 2 or 3, class 3. When the gross weight exceeds 200 pounds, or the container length and width is 48 x 24 inches or more and the weight exceeds 100 pounds, 3 x 4 inch skids, laid flat, shall be applied in accordance with the requirements of the container specification, or if not specified in the specification, in a manner which will adequately support the item and facilitate the use of material handling equipment. Closure and strapping shall be in accordance with the applicable container specification or appendix thereto except that metal strapping shall conform to QQ-S-781, type 1, class B.

5.2.2 Level B.-

5.2.2.1 Consolidation.- A quantity of Radio Set Control Group AN/GRA-39( ), packaged as specified in 5.1, shall be packed as specified in 5.2.1.1.

5.2.2.2 Palletized load.- A quantity of containers, packed as specified in 5.2.2.1, shall be palletized as specified in 5.2.1.2.

5.2.2.3 Less than palletized load.- When quantities per destination are less than a pallet load, the containers packed as specified in 5.2.2.1 shall be reinforced by pressure-sensitive filament tape conforming to PPP-T-97, type IV as specified in the appendix of the box specification. No further packing shall be required.

5.2.3 Level C.-

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5.2.3.1 Consolidation.- A quantity of Radio Set Control Group AN/GRA-39( ), packaged as specified in 5.1, shall be packed as specified in 5.2.1.1, except that the fiberboard boxes shall be class domestic.

5.2.3.2 Palletized load.- A quantity of containers packed as specified in 5.2.3.1, shall be palletized as specified in 5.2.1.2, except that the fiberboard cap shall be class domestic.

5.2.3.3 Less than palletized load.- When quantities per destination are less than a pallet load, the containers packed as specified in 5.2.3.1 shall be used as the shipping containers. No further packing shall be required.

5.3 Marking.- In addition to any special marking required by the contract or order, interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129.

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BILL OF MATERIAL
TABLE VII

Ref No.	ITEM Nomenclature	No. Reqd	Method MIL-P -116	Size - Inches L	W	D	REQUIREMENT Spec No.	T	C	V	G	F	S
1	Radio Set Control Group AN/GRA-39( ) consisting of:	1	111	11-½	9-3/4	9-7/8	(Bare wt. 25 lbs.)						
	Bag CW-598/GRA-39( )	1		All items placed within bag									
	Control, Radio Set C-2328/GRA-39	1											
	Control, Radio Set C-2329/GRA-39	1											
	Handset H-189( )/U	1											
	Sling, Carrying	2											
	Lamps, Spare	2	111										
	Paperboard, Cushioning	2		6	6		PPP-P-291	111					
	Tape	2		6	2		PPP-T-45	111			A		
				Cushion each spare lamp in paperboard and secure wrap with tape.									
2	Pad	1		10½	9	1	PPP-F-320	CF	WR	SW	W5c		
3	Technical Literature	IC-1		10	8	1/8							
	Bag	1		11½	9		MIL-B-117	11	C				
				Place the technical literature in the heat-sealed bag on top of the pad prior to closure.									
4	Box, Fiberboard	1		11-5/8	9-7/8	10	PPP-B-636	CF	WR	SW	W5c		

Lev. A - PACKED WEIGHT
Lev. A - PACKED VOLUME

T - Type V - Variety F - Flute
C - Class G - Grade S - Style

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RADIO SET
AN/GRA-39( )

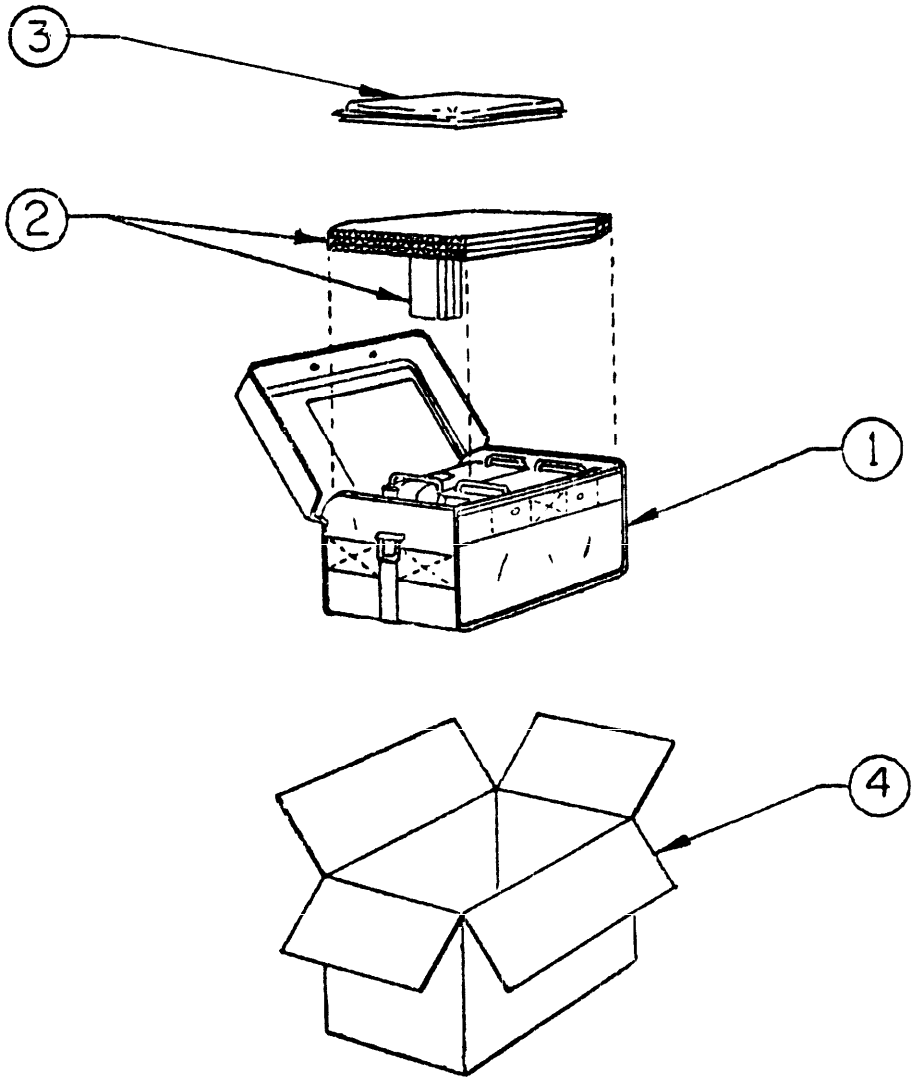


FIGURE 1

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6. NOTES

6.1 Intended use.- Radio Set Control Group AN/GRA-39( ), including Controls C-2328( )/GRA-39 (remote) and C-2329( )/GRA-39 (local), provides means for controlling the operation of tactical Radio Sets AN/VRC-12( ), AN/PRC-25( ), AN/PRC-77, AN/PRC-35 and other similar radio sets. After radio set power is turned on manually, voice communications can be received and transmitted through the radio sets and locally or remotely controlled by the operator up to two (2) miles from the transceiver site by means of telephone field wires and standard switchboards. Facilities for duplex telephone communication between local and remote control operations are provided.

6.2 Ordering data.- Procurement documents should specify the following:

a. Title, number, and date of this specification and any amendment thereto.

b. Type required.

c. First Article pack(s) as follows:

Makeup of pack(s).
Number of each kind of pack to be submitted.
Inspection to be performed thereon.

d. Marking and shipping of samples.

e. Place of final inspection.

f. Technical literature required (see 3.14).

g. Quantity of running spare parts required (see 3.14).

h. Levels of preservation and packaging and packing (see Section 5).

i. If the rough handling and functional tests are required (see 4.16).

j. Dimensional data.- Dimensions of packaging material in the Bill of Material Table are based on the dimensions of the units cited in the table. When the dimensions of the units vary from those cited, the dimensions of the packaging material shall be adjusted accordingly.

6.3 Color.- The color chip furnished by the procuring agency (see 3.4) will be color chip no. X-24087, and may be obtained upon request to the Commanding Officer, US Naval Supply Depot, 5801 Tabor Avenue, Philadelphia, Pa. 19120.

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6.4 Definitions.-

6.4.1 Unit divisions.- The terms used herein for unit divisions conform to Standard MIL-STD-280.

6.4.2 Definitions of inspection terms.- The inspection terms used herein conform to Standard MIL-STD-109. The following are abbreviated definitions selected from those given therein for immediate reference.

6.4.2.1 Inspection.- Inspection is the examination or testing, or both, of supplies to determine compliance with applicable requirements. Sampling is an element of inspection.

6.4.2.2 Examination.- Examination consists of simple, generally nondestructive determinations of compliance, without use of special testing equipment.

6.4.2.3 Testing.- Testing consists of determinations of compliance, using technical means.

6.5 Humidity.- Natural climatic extremes may include 100 percent relative humidity, with condensation, up to 4 hours at a time. This requirement (see 3.7.2) is specified by Standard MIL-STD-210. The moisture-resistance test (see 4.6.2), using Standard MIL-STD-810, is a procedure that gives a practical test approximating the natural conditions.

6.6 Group C inspection.- Approval to ship may be withheld, at the discretion of the government, pending the decision from the Contracting Officer on the adequacy of corrective action (see 4.5.3.2).

6.7 Location of operational inspection and air-seal test.- It is desirable that the operational inspection (4.8.7) and the air-seal test (4.9) be performed at a location that will minimize handling (which might cause damage to the equipment) after these inspections are completed. Any preparation for shipment which would require breaking of the equipment seal should be accomplished prior to the air-seal test so that the seal may remain intact thereafter. It is recommended that the entire lot (including all previously inspected sample units) be sampled and inspected immediately prior to packaging.

6.8 Nomenclature.- The parentheses in the nomenclature will be deleted or replaced by a letter identifying the particular design; for example: C-2328W/GRA-39. The contractor should apply for nomenclature in accordance with the applicable clause in the contract (see 1.1).

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6.9 Verification inspection.- Verification by the government will be limited to the amount deemed necessary to determine compliance with the contract and will be limited in severity to the definitive quality assurance provisions established in this specification and the contract. The amount of verification inspection by the government will be adjusted to make maximum utilization of the contractor's quality control system and the quality history of the product.

Custodian:
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Preparing Activity:
Army-EL

Project No. 5820-A558

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION MIL-R-55154B(EL) RADIO SET CONTROL GROUP AN/GRA-39( )
(UNITS OF)

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?
   ☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

ESC-FM 1068-68

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Source: https://assist.dla.mil -- Downloaded: 2026l0-26T04:14Z
Check the source to verify that this is the currentversion before use.